

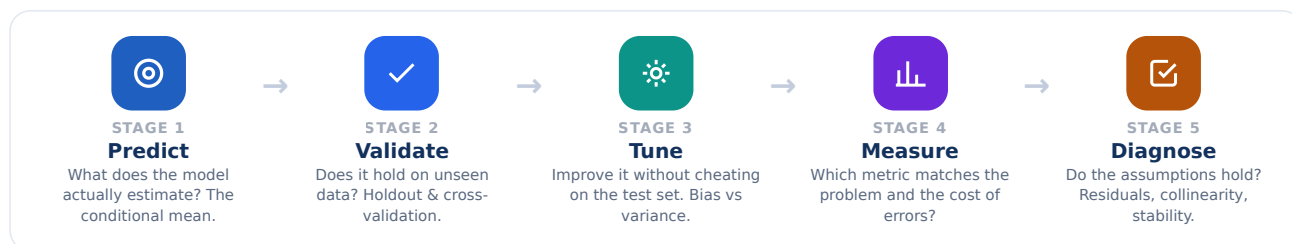
“Would this model still be right on a slab it has never seen before?”

The Evaluation Workflow

From “what does it predict?” to “can I trust it?”

Evaluation is not one number at the end — it is five questions asked in order. Skip one and you can ship a model that looks brilliant and fails in production.

THE FIVE-STAGE WORKFLOW



The one question behind it all

Every stage is a way of answering one thing honestly: **how well does this model do on data it has never seen?** Training performance is a diagnostic, not an achievement.

Fit $\neq$ Generalization

Accuracy $\neq$ Validity

Software ran $\neq$ Answer is right

Three verdicts you're trying to avoid



GOLDEN RULES OF HONEST EVALUATION

- 1 Judge on data the model never saw.** The cross-validation score is the truth; the training score is just a diagnostic.
- 2 Match the metric to the decision,** not to habit. The cost of each error picks the metric.
- 3 Keep one test set untouched** until the very end — tune on cross-validation only.
- 4 A model can predict well and still break assumptions** — read metrics *and* diagnostics.
- 5 Prevent group leakage:** the same slab in train and test inflates every score.
- 6 Never trust a single number** — read fit and generalization together, always.

★ KEY TAKEAWAY

Evaluation is a **sequence of honest questions**, not a final score. Ask all five, in order,

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Let the cost of the error — a missed defect vs. a false alarm — choose the metric.

The Metric Cheat Sheet

Match the metric to the problem — and to the mistake that costs you money.

The first rule of metrics: the metric must match the problem type, what the model predicts, and which error is actually expensive. Pick before you model, not after.

REGRESSION · PREDICTING A NUMBER

METRIC	MEASURES	FORMULA	REACH FOR IT WHEN
MAE	Average error size, all errors equal	$\text{mean} y - \hat{y} $	You want a plain, robust “typical miss.”
MSE	Squared error — punishes big misses	$\text{mean}(y - \hat{y})^2$	Large errors are especially bad.

RMSE	MSE back in the target's own units	$\sqrt{\text{MSE}}$	You want MSE's penalty, readable units.
R²	Share of variance explained (0→1)	$1 - \text{SS}_r / \text{SS}_t$	Communicating fit to non-specialists.

CLASSIFICATION · A CLASS

Accuracy (TP+TN)/all · balanced only	Precision TP/(TP+FP) · false alarms costly
Recall TP/(TP+FN) · a miss is dangerous	F1 balance, imbalanced data
ROC-AUC separability across all thresholds; threshold-free comparison	

The confusion matrix

	Pred +	Pred -
Act +	TP	FN missed
Act -	FP false alarm	TN

Every classification metric is just a ratio of these four counts.

Clustering · no labels

Silhouette — cohesion vs. separation (−1→+1). No ground truth; judge how tight the groups are.

Adjusted Rand Index — agreement with known labels, corrected for chance.

Quantile · a percentile

Pinball loss — asymmetric error at quantile τ ($\tau=0.5 \rightarrow$ MAE; $\tau=0.9$ penalizes under-prediction 9×).

Pseudo R² — fit vs. a null model. McFadden / Koenker-Machado form.

R² and RMSE measure the wrong target here — they assume you predicted the mean.

The costed-error rule

Missing a defect is dangerous → optimize **Recall**

False alarms waste money → optimize **Precision**

Big misses ruin you → prefer **RMSE/MSE**

Imbalanced classes → never trust **Accuracy**

★ KEY TAKEAWAY

The metric must match the **problem type** and the **cost of the mistake**. Choose it before you model.

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